Machine Learning Projects

ML Tasks, Detailed Project Descriptions, Dataset Links and Expected Outcomes

Team No.	Project Title	ML Tasks	Detailed Project Description	Dataset & Link	Expected Outcomes
1	Explainable Student Dropout Risk Prediction Using Machine Learning	Classification, feature selection, explainable ML, imbalanced learning	Develop a model to classify students into dropout, enrolled, and graduate categories. Perform preprocessing, class-imbalance handling, feature selection and compare multiple ML classifiers. Add SHAP/LIME-based explanations to identify factors influencing individual predictions.	**Predict Students' Dropout and Academic Success** Open Dataset https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success	High-performing dropout prediction model; important risk factors; individual prediction explanations; confusion matrix, F1, ROC-AUC and balanced accuracy; early-warning framework.
2	Early Industrial Machine Failure Prediction Using Cost-Sensitive Machine Learning	Binary classification, regression, cost-sensitive learning, feature engineering	Use machine operating parameters such as temperature, torque, rotational speed and tool-wear to predict whether a failure is likely. Assign a higher cost to missed failures and compare conventional and cost-sensitive models.	**AI4I 2020 Predictive Maintenance Dataset** Open Dataset https://archive.ics.uci.edu/dataset/601/ai4i	Machine-failure prediction model; early-warning probability score; reduced false negatives; comparison of conventional vs cost-sensitive ML; important machine parameters.
3	Adaptive Air Quality Prediction Using Machine Learning	Regression, time-series feature engineering, ensemble learning, adaptive learning	Predict pollutant concentration from gas-sensor measurements and environmental variables. Generate lag, rolling-average and trend features. Compare regression and ensemble approaches and evaluate performance under changing conditions.	**Air Quality** Open Dataset https://archive.ics.uci.edu/dataset/360/air+quality	Pollution prediction model; short-term forecasting; important pollutant/sensor factors; MAE, RMSE and R^2 comparison; adaptive prediction strategy.
4	Personalized Obesity Risk Prediction Using Explainable Machine Learning	Multiclass classification, feature selection, explainable ML	Predict obesity categories from eating habits, physical condition and lifestyle-related attributes. Compare classifiers and identify features responsible for individual predictions.	**Estimation of Obesity Levels Based On Eating Habits and Physical Condition** Open Dataset https://archive.ics.uci.edu/dataset/544/estimation+of+obesity+levels+based+on+eating+habits+and+physical+condition	Obesity-level prediction model; personalized risk factors; feature-importance ranking; explainable individual predictions; comparative ML performance.
5	Fairness-Aware Credit Default Prediction Using Machine Learning	Binary classification, fairness analysis, probability calibration	Predict whether a customer is likely to default on credit-card payments. Compare standard models and analyze whether prediction errors differ between demographic groups. Investigate threshold adjustment and probability calibration.	**Default of Credit Card Clients** Open Dataset https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients	Credit-default prediction model; calibrated probabilities; fairness/error comparison; accuracy, precision, recall, F1 and ROC-AUC; fairness-aware strategy.
6	Early Corporate Bankruptcy Detection Using Imbalanced Machine Learning	Imbalanced classification, SMOTE, ensemble learning, cost-sensitive learning	Build an early-warning system for financially distressed companies. Compare class weighting, SMOTE/ADASYN and ensemble approaches, optimizing for recall of bankruptcy cases rather than raw accuracy.	**Polish Companies Bankruptcy** Open Dataset https://archive.ics.uci.edu/dataset/365/polish+companies+bankruptcy	Bankruptcy-risk classifier; improved minority-class recall; imbalance-method comparison; feature importance; early financial-risk warning.
7	Electricity Consumption Anomaly Detection Using Unsupervised Machine Learning	Clustering, anomaly detection, dimensionality reduction	Analyze electricity-consumption patterns without manually labelled anomalies. Create consumption profiles and use Isolation Forest, Local Outlier Factor, clustering or related unsupervised methods to identify unusual behavior.	**ElectricityLoadDiagrams20112014** Open Dataset https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014	Normal consumption profiles; automatically detected anomalies; abnormal-period visualization; anomaly-method comparison; anomaly scoring mechanism.
8	Intelligent Customer Segmentation Using RFM and Machine Learning Clustering	Clustering, feature engineering, dimensionality reduction	Calculate Recency, Frequency and Monetary (RFM) features from transaction data. Apply K-Means, hierarchical clustering and DBSCAN to discover customer groups and identify high-value, inactive and potential customers.	**Online Retail** Open Dataset https://archive.ics.uci.edu/dataset/352/online+retail	Customer segments; RFM profiles; optimal cluster analysis; silhouette-score comparison; actionable customer profiles.
9	Indoor Position Prediction Using Machine Learning from BLE Signal Fingerprints	Multiclass classification, feature engineering, noisy-data handling	Predict indoor location from Bluetooth signal-strength fingerprints. Investigate noise, missing signals and feature selection. Compare KNN, Random Forest, SVM and ensemble models.	**BLE RSSI Dataset for Indoor Localization and Navigation** Open Dataset https://archive.ics.uci.edu/dataset/497/ble+rssi+dataset+for+indoor+localization+and+navigation	Indoor location prediction; robustness to noise; location-wise accuracy; signal-feature importance; average localization error.
10	Human Activity Recognition Using Sensor-Based Machine Learning	Multiclass classification, feature selection, sensor fusion	Classify human activities using smartphone sensor features. Compare feature subsets and investigate whether feature selection can maintain performance while reducing computational complexity.	**WISDM Smartphone and Smartwatch Activity and Biometrics Dataset** Open Dataset https://archive.ics.uci.edu/dataset/507/wisdm+smartphone+and+smartwatch+activity+and+biometrics+dataset	Activity recognition model; reduced feature set; accuracy vs computational-cost comparison; confusion analysis; lightweight ML model.
11	Automated Feature Selection for High-Dimensional Machine Learning	Feature selection, dimensionality reduction, classification	Use the Madelon benchmark to identify informative features from a high-dimensional dataset containing irrelevant and redundant variables. Compare filter, wrapper and embedded feature-selection strategies.	**Madelon** Open Dataset https://archive.ics.uci.edu/dataset/171/madelon	Reduced feature subset; informative-feature identification; comparable/improved classification; reduced training time; feature-selection methodology.
12	Dynamic Energy Demand Forecasting Using Ensemble Machine Learning	Regression, time-series forecasting, ensemble learning	Predict future electricity/energy demand using historical consumption and time-related features. Generate lag and rolling-window features and compare Random Forest, Gradient Boosting, XGBoost and stacking.	**ElectricityLoadDiagrams20112014** Open Dataset https://archive.ics.uci.edu/dataset/321/electricityloaddiagrams20112014	Short-term energy forecast; demand trends; ensemble prediction model; MAE/RMSE/MAPE comparison; peak-demand prediction.
13	Suspicious Company Detection Using Machine Learning-Based Anomaly Detection	Anomaly detection, classification, clustering	Develop an ML system to identify potentially abnormal financial records. Compare supervised classification with unsupervised anomaly detection and investigate combining anomaly scores with classification.	**Audit Data** Open Dataset https://archive.ics.uci.edu/dataset/475/audit+data	Suspicious-record identification; anomaly score; supervised vs unsupervised comparison; precision-recall analysis; high-risk ranking.
14	Early Student Performance Prediction Using Machine Learning	Classification, regression, feature engineering, explainability	Predict student academic performance using demographic, academic and behavioral attributes. Develop classification and regression versions, such as at-risk/not-at-risk and predicted final grade.	**Student Performance** Open Dataset https://archive.ics.uci.edu/dataset/320/student+performance	Early performance prediction; at-risk identification; important performance factors; grade prediction; intervention-oriented ML insights.
15	Rice Variety Classification Using Lightweight Machine Learning	Multiclass classification, feature selection, model compression	Classify rice varieties using physical/morphological features. Compare classifiers and reduce the number of features while preserving performance, targeting	**Rice (Cammeo and Osmancik)** Open Dataset https://archive.ics.uci.edu/dataset/545/rice+cammeo+and+osmancik	Rice-variety classifier; reduced-feature model; accuracy-performance trade-off; feature importance; lightweight deployment-

			resource-constrained environments.		ready model.
16	Automated Dry Bean Variety Classification Using Explainable Machine Learning	Multiclass classification, feature selection, explainability	Classify seven dry-bean varieties using geometric and shape features. Compare ML models and determine which morphological characteristics distinguish each variety.	**Dry Bean** Open Dataset https://archive.ics.uci.edu/dataset/602/dry+bean	Seven-class bean classifier; important morphological features; class-wise performance; explainable classification; reduced-feature model.
17	Multi-Stage Air Pollution Prediction Using Feature Selection and Ensemble Learning	Regression, feature selection, ensemble learning, model comparison	First identify informative sensor variables, then build multiple regression models and combine predictions using ensemble/stacking. Compare the framework against individual baseline models.	**Air Quality** Open Dataset https://archive.ics.uci.edu/dataset/360/air+quality	Selected sensor subset; improved pollution prediction; ensemble/stacking model; error reduction; sensor contribution analysis.
18	Forest Cover Type Prediction Using Large-Scale Machine Learning	Multiclass classification, feature selection, ensemble learning	Predict forest-cover categories from cartographic and environmental features. Investigate feature importance, class-wise performance, ensemble models and the effect of training-data size.	**Covertype** Open Dataset https://archive.ics.uci.edu/dataset/31/covertype	Forest-cover classifier; feature-importance ranking; class-wise analysis; scalable ML pipeline; performance vs training-data-size study.

Recommended Common ML Workflow

1. Dataset Understanding
2. Data Cleaning and Missing-Value Treatment
3. Exploratory Data Analysis
4. Feature Engineering
5. Feature Selection / Dimensionality Reduction
6. Baseline ML Model
7. Proposed / Improved ML Model
8. Hyperparameter Optimization
9. Model Evaluation
10. Explainability / Error Analysis
11. Comparative Analysis
12. Final ML Prediction System

Suggested ML Algorithms

ML Task	Suggested Algorithms
Classification	Logistic Regression, Decision Tree, Random Forest, SVM, KNN, XGBoost
Regression	Linear Regression, Random Forest Regressor, Gradient Boosting, XGBoost
Clustering	K-Means, Hierarchical Clustering, DBSCAN
Anomaly Detection	Isolation Forest, Local Outlier Factor, One-Class SVM
Feature Selection	Mutual Information, RFE, LASSO, Random Forest Importance
Imbalanced Learning	Class Weighting, SMOTE, ADASYN, Cost-Sensitive Learning
Ensemble Learning	Voting, Bagging, Boosting, Stacking
Explainable ML	SHAP, LIME, Permutation Importance
Model Optimization	Grid Search, Random Search, Bayesian Optimization

Minimum Expected Project Deliverables

1. Baseline model establishing initial performance.
2. Improved/proposed ML model demonstrating the project's innovation.
3. Comparative evaluation using at least 3–5 ML algorithms.
4. Feature analysis identifying influential variables.
5. Error analysis explaining model failures and limitations.

6. Final prediction interface/report demonstrating the model on unseen data.